Clinical Trial Eligibility Compiler

Eligibility criteria compiled into executable predicates, evaluated deterministically, reported as dates rather than labels.

Scope non-small cell lung cancer · English · local web UI · 500 recruiting trials

500
trials

13,202
criteria

15,360
predicates

0
model calls

Source: ClinicalTrials.gov API v2 · snapshot 14 Aug 2026 · every trial links to its registry record

THE PROBLEM

Both sides of the market are starved at once

< 5%

of adult cancer patients enrol in a trial

20–40%

of cancer trials miss enrolment targets and terminate early

100,000s

of registered trials to search

Why patients do not enrol

The mechanism. Eligibility criteria are free-text prose: negation, lab thresholds with units, temporal washout windows, prior therapy line counts, biomarker requirements — written for a human coordinator to read by hand.

WHAT ONE CRITERION CONTAINS

Four kinds of arithmetic in a single sentence

“ALT and AST ≤ 3× upper limit of normal, or ≤ 5× ULN in the presence of liver metastases”

- **Two analytes, one threshold** — a conjunction — a passing AST must not mask a failing ALT
- **Relative, not absolute** — 3× ULN is meaningless without the institution's own limit. 37% of lab criteria are ULN-relative.
- **A conditional variant** — which bound applies depends on a separate clinical fact about the patient
- **Units that will not match** — the report says 1500 cells/mm^3; the protocol says 1.5×10^9/L

Criterion text verbatim from NCT05283330.

PRIOR WORK

Reasoning-based trial matching already exists

TrialGPT

NIH, 2024

Criterion-level eligibility with natural-language explanations and sentence-level evidence, aggregated to a ranked list.

87.3% criterion accuracy
NDCG@10 0.7252 · P@10 0.6724

Leach et al.

Kentucky, arXiv 2512.08026

Four eligibility classes including “could be eligible in future”. Source citation and a web interface listed as future work.

Evaluated on synthetic records
No ground truth · ~950 LLM calls/patient

PRISM, TrialMatchAI

Deep 6 · Triomics · Tempus

Commercial and academic systems operating at far greater breadth than this project attempts.

Not competing on breadth

This project does not claim to have invented reasoning-based trial matching. A claim of conceptual novelty would not survive question and answer.

THE CLAIM

What is actually new here

Reasoning-based trial matchers now produce temporal eligibility labels, but they **route numeric and date logic through the language model** and none have been quantitatively validated on criterion-level ground truth.

Compiles	eligibility criteria into executable predicates
Evaluates	them deterministically, in Python, with zero model calls
Produces	specific eligibility dates, not categorical labels
Reports	the first measured comparison of compiled vs prompted numeric evaluation

EVIDENCE

The difference, measured

	A prompted matcher	This system
Determinism	Sampled output. The same patient re-run can change a verdict, and no system on the previous slide reports run-to-run variance at all.	20 runs of the full 500-trial query, one SHA-256 over every field but the clock. ● **1 distinct output hash / 20 runs**
Temporal answer	A category — “could be eligible in future”. The coordinator still has to work out when, from the note.	A calendar date, computed from an anchor date read out of the note and shown beside it. ● **16 Sep 2026 · 30 Sep 2026**
Provenance	Evidence quoted from the trial text, generated by the same pass that produced the answer.	Character offsets into the patient record, emitted by the evaluator, not by a model. ● **75.6% of verdicts click to their source sentence**
Missing data	Answers regardless. A note that never mentions the field still yields a verdict rather than a question.	Abstains, and names the test that would settle it — a gap becomes a work order. ● **10.6% abstained · top 4 tests resolve 340 criteria**

Where this loses. TREC 2022 NDCG@10 0.3600 against TrialGPT’s published 0.7252 — those notes carry no lab values and no dates, so no washout predicate can fire. And the compiled-vs-prompted numeric head-to-head has not been run.

Measured by eval/differentiation.py over the 500-trial corpus and one persona. Determinism excludes the timing block, which varies by construction.

ARCHITECTURE

Do the hard work once, offline

OFFLINE · run once, cached to disk

- **Pull** — 500 recruiting NSCLC trials, ClinicalTrials.gov v2
- **Split** — eligibility blocks → 13,202 criteria
- **Compile** — criteria → 15,360 typed predicates
- **Persist** — predicate cache on disk

the entire technical contribution

ONLINE · per patient query

- **Extract** — report → patient record, with a character span on every field
- **Retrieve** — candidate trials
- **Evaluate** — predicates in Python — no model calls
- **Aggregate** — to four states, with dates

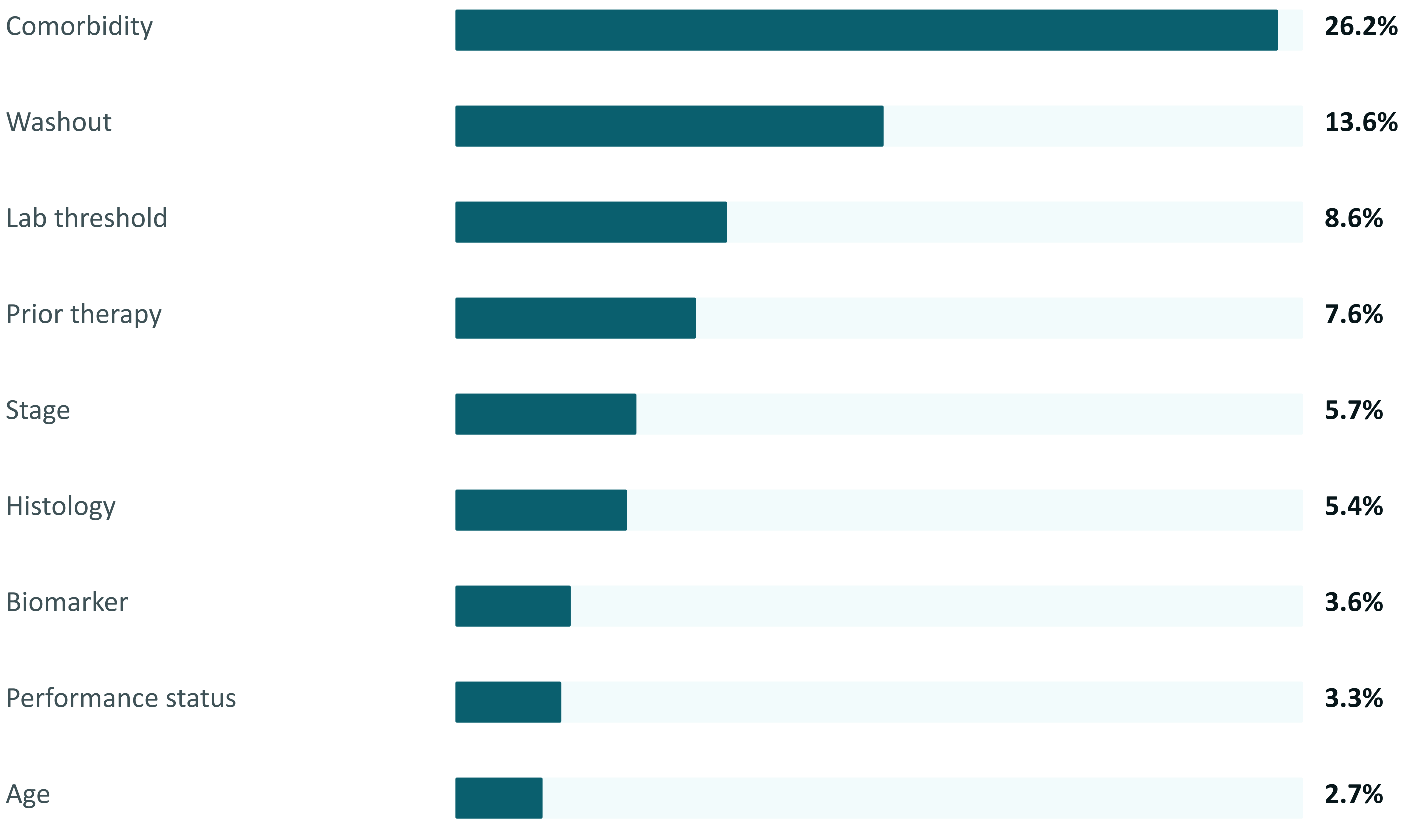
65 ms median for 500 trials · 50 warm runs

The LLM performs extraction only. It never performs a comparison, a unit conversion, or date arithmetic — every numeric and temporal verdict comes from Python.

METRIC 9.1

How much of eligibility prose is deterministic?

Criteria compiled, by predicate type (% of all criteria)



39.7% stays free text and is routed to the model

60.3%

of all criteria compile to a typed predicate

hand-read ceiling is 70.4% — the compiler reaches 86% of what the schema can cover

Per-type F1, scored against 695 hand-read criteria

Age	.978
Performance status	.964
Lab threshold	.951
Washout	.926
Histology	.892
Comorbidity	.877
Stage / Biomarker	.862
Prior therapy	.832

Nobody has published this number. It quantifies how much of trial eligibility is deterministic — reported as a finding, not a system statistic.

METRIC 9.2 · THE HEADLINE

Compiled numeric criterion accuracy

100%

on 32 held-out numeric criteria

written after the fixes, run exactly once

146

hand-labelled criterion instances

every criterion string verbatim from the corpus, with its NCT ID

8

real bugs the eval found before any chart existed

none of which 368 unit tests had caught

Why a held-out split, and why the 146-case number is not quoted

The development set found eight bugs, which were then fixed. That makes its post-fix score a regression suite, not an estimate — quoting it would be tuning on the test set. The held-out cases were written afterwards and run once. 100% of 32 has a wide interval (~89–100%); the honest statement is “no errors on 32 held-out numeric criteria”.

Clean on every category the thesis depends on

- unit conversion
- ULN-relative bounds
- calendar months
- month-end clamping
- leap years
- molar conversion
- strict vs inclusive
- exclusion polarity

Prompted baseline: harness built and fairness-audited — identical input string, all four verdicts defined, free to reason, temperature 0. Awaiting an API key; report whatever it yields.

WHY THE EVAL WAS WORTH BUILDING

Eight bugs, none caught by 368 unit tests

Exclusion polarity never applied	Every exclusion criterion of six types evaluated backwards. A patient with a normal QTc was blocked; one with a prolonged QTc passed. This was live in the UI.
Floating point on ULN bounds	1.5 × 1.2 is 1.7999999999999998, so a patient at exactly 1.8 failed a ≤ 1.5× ULN bound — silently, precisely on the boundary the protocol cares about.
A conjunction compiled as a disjunction	“AST and ALT ≤ 3× ULN” shared one predicate group, so a passing AST masked a failing ALT.
QTc in seconds never converted	0.485 s read as 0.485 ms.

Also: a documented “never received” ignored · a parent drug class emitted alongside its child · an abbreviation counted as a second analyte · two names for one concept across two modules. Three of my own gold labels were wrong and are corrected in place, with the reasoning recorded.

METRIC 9.3 · COST AND LATENCY

At query time this is Python, not inference

0

model calls per patient query

65 ms

median, to match one patient against all 500 trials

$0.00

marginal cost per query

Comparison of architecture, not of implementations

Kentucky pipeline

~950 LLM calls per patient

one model call per candidate trial

This system

0 LLM calls per patient

predicates compiled once, then evaluated in Python

Extraction is one call per patient and is cached, so the demo never touches the network.

THE OUTPUT

Four states, and a date instead of a label

42

Eligible now

every criterion satisfied

2

Eligible on a date

blocked only by a washout; reports the date and the anchor

35

Undetermined

no hard block, but data is missing — lists exactly which

421

Blocked

an immutable criterion fails; one-line reason

The date is computed, not guessed

last dose 19 Aug 2026 + 4-week washout → eligible 16 Sep 2026

Calendar arithmetic, not 30-day months: six months from 31 Aug is 28 Feb, and 31 Jan + 1 month clamps to 28 Feb — 29 Feb in a leap year. If the anchor date is absent the verdict is Undetermined, never an estimate.

Missing-data panel

Order 4 tests to resolve 340 criteria across 220 trials.

Silence is not absence. A condition absent from the record is Undetermined, never “negative” — which is what makes this list actionable.

Ranking on TREC Clinical Trials 2022

NDCG@10 · 50 topics, 12,682 candidate trials

The finding worth keeping

The compiled predicates are worthless alone (0.0147), yet adding them to BM25 lifts NDCG@10 by +19% and P@10 by +32% relative. BM25 separates relevant from irrelevant; predicates separate eligible from excluded. Neither does the other's job.

We are at roughly half of TrialGPT

Four reasons, all of which a knowledgeable judge would raise unprompted:

1. TrialGPT reranks with an LLM at criterion level. This uses zero model calls — a different class of system.
2. The published NDCG gain convention is unconfirmed; both linear and exponential are reported.
3. Ranking runs over TrialGPT's own pool, so only the ranker is compared.
4. TREC notes are ~600 characters and span all of medicine. Extraction recovers zero stage, zero ECOG, zero biomarkers and zero dated therapies — so not one washout predicate can fire.

α was fixed before the run. A sweep shows α = 0.3 scores marginally higher; that is not quoted, because choosing it after seeing the test set is the error the held-out split exists to avoid.

LIMITATIONS

Stated before being asked

Extraction errors propagate downstream

Mitigated by span-level provenance on every field, and by abstaining to Undetermined rather than guessing.

Recruiting status is frequently stale

Sites listed as recruiting are often closed. Fixed snapshot, 14 Aug 2026; every trial links to its ClinicalTrials.gov record for live status.

Computed dates depend on a last-dose date

Often stated imprecisely, or not at all. Where it is stated the date is exact calendar arithmetic; where it is absent the verdict is Undetermined, never an estimate.

TREC notes are synthetic and thin

Cleaner than real clinical text, and missing the structured fields this system runs on.

This is coordinator and clinician decision support

Not patient-facing advice. It makes no treatment recommendation.

NSCLC only

Trial density varies several-fold across cancer types. These results do not transfer without re-measurement.

What is defensible

60.3% of eligibility criteria compile to executable predicates — a number nobody has published

0 errors on 32 held-out numeric criteria, including unit conversion, leap years and month-end clamping

0 model calls per patient query. 500 trials matched in 65 ms

8 bugs found by the eval before it produced a single chart

Every figure in this deck is reproducible from the repository: python3 eval/compile_rate.py · numeric_accuracy.py --holdout · ranking_benchmark.py